CLINICAL · APPROACH TO

Approach to a blocked or full ear

"My ear feels blocked" is one of the commonest things patients say, and the cause can lie anywhere from the canal to the cochlea. Most blocked ears are wax, a middle ear effusion or a sluggish Eustachian tube. The ones not to miss are the sudden sensorineural hearing loss, which feels like a blocked ear and needs steroids within days, the unilateral effusion in an adult that hides a nasopharyngeal tumour, and the early Ménière's disease that announces itself as fullness with low-frequency loss.

AT A GLANCE

- Otoscopy and tympanometry before anything else.
- Sudden sensorineural loss presents as a blocked ear: test the hearing that day.
- Unilateral effusion in an adult: examine the nasopharynx.
- Autophony: patulous tube or superior canal dehiscence.

1 CHAPTER ONE Where is the block?

Canal, middle ear, tube or cochlea.

1.1 Four places

The sensation of fullness has four main sources. **The canal** contributes wax, debris, a foreign body, otitis externa or exostoses. **The middle ear** contributes an effusion, a retracted or perforated drum, haemotympanum or, rarely, a mass. **The Eustachian tube** fails to open (obstructive dysfunction) or stays open (patulous tube). **The inner ear** produces fullness in Ménière's disease, sudden sensorineural loss, superior canal dehiscence and other third-window lesions, and with raised intracranial pressure. Temporomandibular joint disorders and anxiety can also make an ear feel full.

DO NOT MISS

- **Sudden hearing loss with fullness:** test the hearing that day. A sensorineural loss of 30 dB or more over three frequencies within 72 hours needs steroids now and MRI later.

- **A unilateral middle ear effusion in an adult:** examine and, if in doubt, biopsy the nasopharynx.
- **Fullness with low-frequency loss and episodic vertigo:** Ménière's disease.
- **Fullness with pulsatile tinnitus, headache and visual blurring:** idiopathic intracranial hypertension.

2 CHAPTER TWO History

Small differences matter.

2.1 The questions

Ask whether the fullness came on suddenly or gradually, and whether it is constant or comes and goes. A *sudden* onset, especially on waking, with muffled hearing and tinnitus, is a sudden sensorineural loss until tested. Fullness after a cold, a flight or a dive suggests Eustachian tube dysfunction or barotrauma.

Ask whether the patient can "pop" the ear and whether that relieves it. Obstructive tube dysfunction improves when the ear pops; a patulous tube makes the patient hear their own voice and breathing loudly (*autophony*), improves when lying down or with a cold, and is common after weight loss and in pregnancy. Autophony with hearing one's own eye movements or footsteps, or vertigo with loud noise or straining, points to superior canal dehiscence.

Ask about episodic vertigo, fluctuating hearing and roaring tinnitus (Ménière's disease), nasal blockage, rhinitis and reflux (tube dysfunction), jaw pain and clicking (temporomandibular disorder), and in adults about epistaxis, nasal blockage, neck lumps and ethnicity (nasopharyngeal carcinoma).

3 CHAPTER THREE Examination and tests

Look, then measure.

3.1 Examination

Otoscopy shows wax, otitis externa, a perforation, a dull retracted drum with an air–fluid level or bubbles, or a blue drum of haemotympanum. Watch the drum during deep breathing: in a patulous tube it moves in and out with respiration. Tuning forks are essential, because they separate a conductive from a sensorineural loss at the bedside within minutes. With Weber to the blocked ear and Rinne negative, the loss is conductive; with Weber away from the blocked ear and Rinne positive, it is sensorineural.

Examine the nose and nasopharynx with an endoscope, and palpate the temporomandibular joints and the neck.

3.2 Tests

Pure tone audiometry is the key test. A conductive loss points to the canal or middle ear, a low-frequency sensorineural loss to hydrops, and a sudden flat or high-frequency sensorineural loss to sudden sensorineural hearing loss. **Tympanometry** shows a flat type B trace with normal canal volume in an effusion, a negative-pressure type C trace in tube dysfunction, and a normal type A trace in inner ear causes, with a large volume if the drum is perforated. Respiratory variation on tympanometry supports a patulous tube.

The **seven-item Eustachian Tube Dysfunction Questionnaire (ETDQ-7)** scores symptoms such as pressure, pain, a clogged feeling, symptoms with colds, popping, tinnitus and muffled hearing, and is used to select patients for treatment. **Tubomanometry** measures how the tube opens under nasopharyngeal pressure in specialist centres. **MRI** is used for an asymmetric sensorineural loss and to examine the nasopharynx, and **CT** for suspected dehiscence.

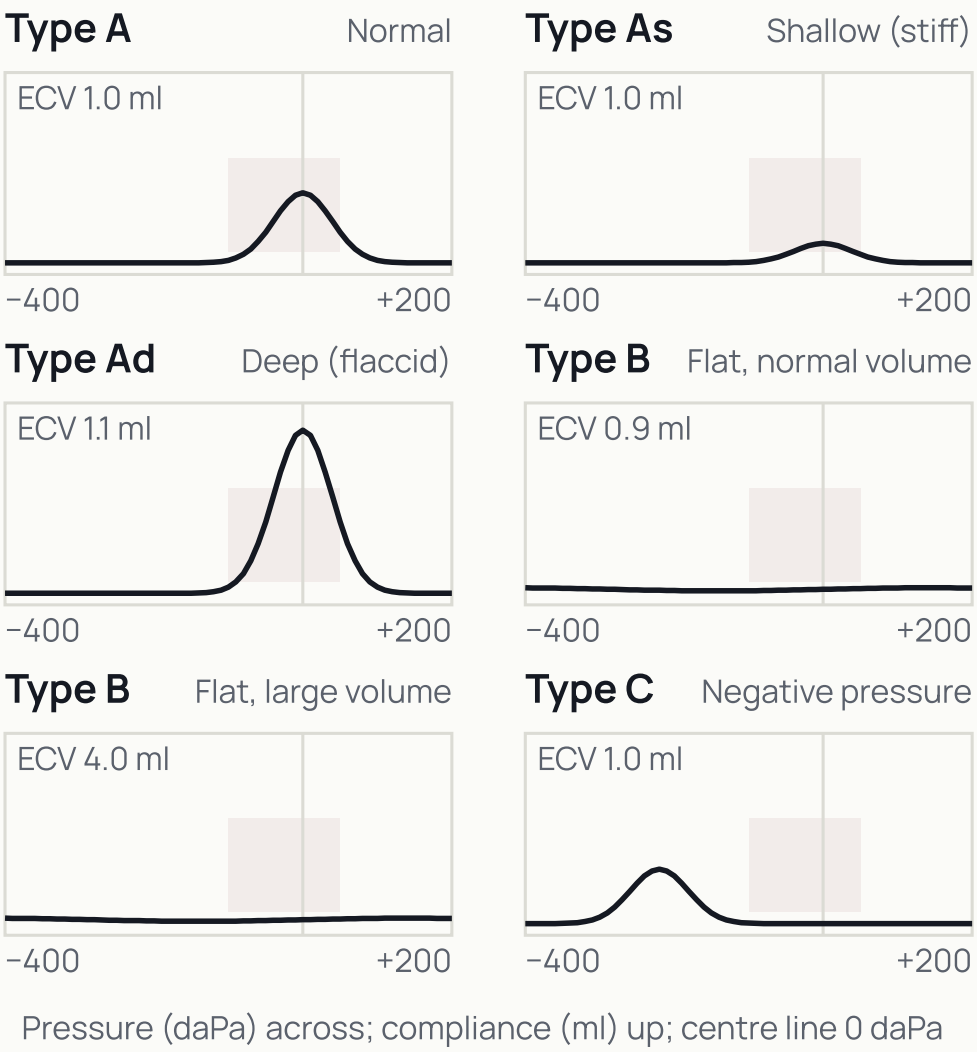


FIGURE 1 Tympanograms in the blocked ear: B with normal volume in an effusion, C in tube dysfunction, A in inner ear causes.

Drawn for ENT Atlas

4 CHAPTER FOUR Causes in turn

What to do for each.

4.1 Wax and the canal

Wax is the commonest cause. It is removed by microsuction, softening drops or irrigation, provided there is no perforation, previous surgery or infection. Otitis externa is treated as such.

4.2 Eustachian tube dysfunction

Obstructive dysfunction is usually *mucosal inflammation* in the cartilaginous tube, from allergic rhinitis, chronic rhinosinusitis, laryngopharyngeal reflux and smoke; in children it is adenoidal hypertrophy and infection, with cleft palate as an important cause. It affects about 1% of adults, and around 40% of children have a temporary episode before age ten.

Management starts with autoinflation (Valsalva or a nasal balloon) and treatment of the nose with saline and a steroid spray. Ventilation tubes treat persistent effusion and retraction. *Balloon dilation Eustachian tuboplasty* is an option for chronic obstructive and baro-challenged dysfunction: a balloon passed through the nose under endoscopic vision is inflated in the cartilaginous tube to about 10 bar for two minutes. Patients are selected by symptoms (ETDQ-7) and testing.

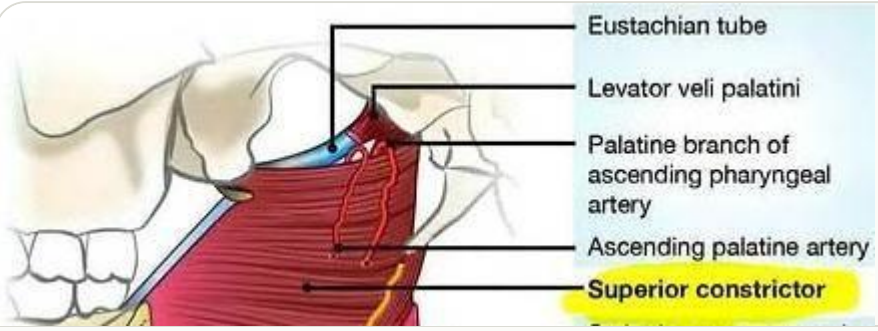


FIGURE 2 The Eustachian tube with the levator and tensor veli palatini, which open it on swallowing.

Anatomy atlas · ENT Steps p. 204

4.3 Patulous Eustachian tube

A patulous tube causes autophony, hearing one's own breathing, and fullness. It is linked to weight loss, pregnancy, hormones, radiotherapy and neuromuscular disease, and some patients "sniff" to close the tube, which can lead to retraction and cholesteatoma. Treatment is explanation, hydration, saline nasal drops and lying down, with augmentation or reconstruction of the tube for disabling cases.

4.4 Middle ear effusion in adults

An adult with a persistent unilateral middle ear effusion needs the nasopharynx examined with an endoscope, and an MRI or biopsy if anything looks abnormal, to

exclude *nasopharyngeal carcinoma*, especially in patients from endemic areas. Other causes are tube dysfunction after a cold, barotrauma, radiotherapy and sinonasal disease. Ventilation tubes relieve persistent effusions once the nasopharynx is clear.



FIGURE 3 Middle ear effusion: bubbles behind the drum on Valsalva.
Dhingra, fig. 10.2

4.5 The inner ear

Sudden sensorineural hearing loss often presents as a blocked ear with tinnitus; an audiogram confirms it and treatment with oral or intratympanic steroids begins at once, followed by MRI. **Ménière's disease** produces fluctuating fullness, roaring tinnitus and low-frequency loss before or during vertigo attacks. **Superior canal dehiscence** produces fullness, autophony, sound- or pressure-induced vertigo and an apparent conductive loss with normal tympanometry and preserved reflexes. **Raised intracranial pressure**, typically idiopathic intracranial hypertension, produces fullness with pulsatile tinnitus and headache.

THE BLOCKED OR FULL EAR

1 Otoscopy

→ **wax or debris**

remove and re-test

→ **effusion, retraction or haemotympanum**
go to step 3

→ **normal drum**
go to step 2

2 Tuning forks and audiogram today

→ **sensorineural loss, sudden**
steroids now, MRI

→ **low-frequency sensorineural, fluctuating, with vertigo**
Ménière's pathway

→ **normal hearing**
step 4

3 Effusion or negative pressure

→ **adult, unilateral, persistent**
endoscope the nasopharynx, biopsy if abnormal

→ **after a cold or flight**
autoinflation, nasal treatment, review

→ **persistent with hearing loss**
ventilation tube

4 Normal hearing and drum

→ **autophony, better lying down**
patulous tube

→ **popping and pressure with colds**
obstructive tube dysfunction — ETDQ-7, consider balloon dilation

→ **jaw clicking and tenderness**
temporomandibular disorder

→ **sound or pressure vertigo**
CT for dehiscence

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **The four sources of fullness;** tuning forks and tympanometry to separate them.
- **Sudden sensorineural loss** presenting as a blocked ear.
- **The adult unilateral effusion** and the nasopharynx.
- **Obstructive versus patulous tube dysfunction:** symptoms, tests, treatment, balloon dilation.
- **Third-window disorders** that mimic conductive loss.

5.2 Common pitfalls

COMMON PITFALLS

- Telling a patient with sudden sensorineural loss that "it's just wax" or "Eustachian tube".
- Inserting a grommet in an adult unilateral effusion without examining the nasopharynx.
- Treating a patulous tube with decongestants, which make it worse.

Written for this app from Oxford Textbook of Otolaryngology ch. 9, 12, Oxford Specialist Handbook ch. 3, Oxford Handbook of ENT ch. 3, Dhingra ch. 5. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

PHARYNX, TONSILS & OESOPHAGUS · TUMOURS

Nasopharyngeal Carcinoma

IMAGING FINDINGS

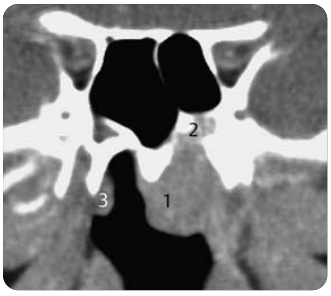
- Asymmetric mass in the fossa of Rosenmüller
- Parapharyngeal and skull base spread; clival marrow replaced on T1
- Retropharyngeal and level II and V nodes

TOP DIFFERENTIALS

- Adenoidal lymphoid hyperplasia (symmetric, striated)
- Lymphoma
- Minor salivary gland tumour

CHECKLIST

- MRI for staging: skull base, cavernous sinus, perineural spread
- Middle ear effusion from Eustachian tube obstruction



CT, coronal with intravenous contrast enhancement. Note the soft-tissue mass in the roof of the nasopharynx of the left side (1), with invasion in the…



Coronal NECT of paranasal sinuses, performed in a 62-year-old Asian man for evaluation of nasal stuffiness and epistaxis, demonstrates an…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 484.

PHARYNX, TONSILS & OESOPHAGUS · INFECTION & INFLAMMATION

Adenoid Hypertrophy

IMAGING FINDINGS

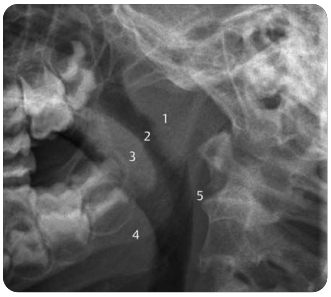
- Symmetric soft tissue in the nasopharyngeal roof and posterior wall
- Striated pattern on T2 with uniform enhancement
- Narrowed nasopharyngeal airway on a lateral film

TOP DIFFERENTIALS

- Nasopharyngeal carcinoma or lymphoma in an adult
- Thornwaldt cyst
- Juvenile angiofibroma

CHECKLIST

- In adults, asymmetric or bulky adenoids need endoscopy
- Middle ear effusions



Plain radiograph for adenoid evaluation.

Dunnebier, Imaging for Otolaryngologists, Fig. 7.2

INNER EAR & IAC · CONGENITAL

Vestibular Aqueduct, Enlargements

IMAGING FINDINGS

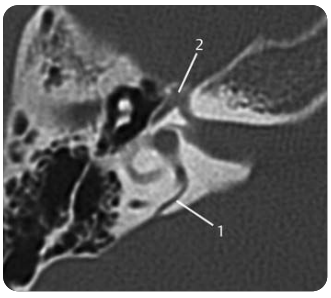
- Vestibular aqueduct wider than about 1 mm at the midpoint or 2 mm at the operculum
- Large endolymphatic sac on T2
- Often with IP-II cochlear deficiency

TOP DIFFERENTIALS

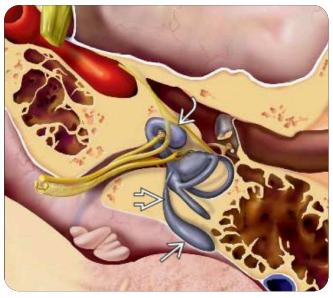
- Normal aqueduct (compare with the posterior canal)
- IP-I cystic cochleovestibular malformation

CHECKLIST

- Look at both ears: often bilateral
- Pendred syndrome association



CT, axial. The normal vestibular aqueduct (1) in its passage from the vestibule to its junction with sinuses in the region of the sigmoid sinus. Ther…



In left inner ear, large endolymphatic sac epidural and intraosseous components are shown. Cochlea is malformed, with absent septation…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1104.